PILOT TRAINING, PROCEDURES, AND CHECKLISTS

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https://simpleflying.com/level-d-simulator/

MITIGATIONS

Individual

Attention & Monitoring

Stress & Decisions

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

CRM

Fatigue

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

Training

Checklists & Procedures

System Design

Displays, Alerts, Warnings

- Controls (soft or hard), "Usability" (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Automation

Operational Risk

Safety 1 and Safety 2

Threat and Error Management

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)

TOPICS

- Pilot training
- Procedures
- Checklists
- Continuing the Flying vs. Driving discussion

PILOT TRAINING – CONCEPTS AND VOCABULARY

- Training programs are covered under different FAA regulations
- Title 14 Code of Federal Regulations (CFR)
  - **Part 61** regulates the certification of pilots
  - **Part 141** regulates pilot schools, more formal program with FAA-approved standard curriculum
  - **Part 142** is a “flight center” with simulator training for professional pilots (no airplanes), emphasizes instrument flying skills, checklists, various types of failures, etc.
  - **Advanced Qualification Program (AQP)** a voluntary alternative method used by most major airlines in the United States
  - **Ab Initio** training programs are run by some airlines (e.g., Lufthansa)

AQP TRAINING

- Formal mapping of training goals and evaluation criteria to simulator sessions.
- Responsive to changes, continuous improvement, evidence-based training scenarios.
- Based upon use of Line Operational Simulation (LOS)
  - LOS used for training are called Line Oriented Flight **Training** (LOFT)
  - LOS used for evaluation are called Line Operational **Evaluation** (LOE)
- Evaluating CRM
  - For example, the ATSB report on QF 32 refers to Salas et al. (1999) and lists a table of observable CRM skills (see lecture slides from 1/28/26)
  - See also Pettitt, M. & Dunlap, J. Skills Development and Assessment in the AQP Environment. Skills Development in AQP (faa.gov)

Salas, E., Prince, C., Bowers, C., Stout, R., Oser, R.L., & Cannon-Bowers, J.A. (1999). A methodology for enhancing crew resource management training. *Human Factors, 41,* p.163.

ON THE JOB PILOT TRAINING

- Promotion sequence within an airline – going from FO to CA position
- Initial operating experience (IOE) – first time with passengers and a check airman
- Types of pilots within an airline
  - Line pilot
  - Check airman
  - Simulator Instructor
  - Technical pilot
  - Management pilot

ESTABLISHING AND MAINTAINING CULTURE NORMS

- Philosophy, Policies, Procedures ➔ Practice
- (Company) Philosophy
  - Overarching view of how to conduct business (flight operations)
  - Related to company culture
  - Example: How much discretion the individual pilot has.
    - Can the FO call for a rejected takeoff when they are the PF or does the CA have to call for it regardless?
  - Example: When to use Automation?
    - Use it as often as possible or is the pilot allowed the discretion to fly manually?

PROCEDURES

- As opposed to individual "pilot technique"
- Standard Operating Procedures – improve standardization of procedures, but can vary between companies/operators
- Procedures unambiguously specify
  - What the task is
  - When the task is done (time and sequence)
  - By whom
  - How (actions)
  - The sequence of actions
  - The feedback (callout, indicator)
- The result is a task that is carried out efficiently, logically, in an error-resistant manner. The procedure should also promote coordination.
- Examples: approach procedure, landing procedure, departure procedure

Degani, A. & Weiner, E.L. (1994) *On the Design of Flight Deck Procedures*. NASA Contractor Report 177642. https://ntrs.nasa.gov/citations/19940029437

PROCEDURE EXAMPLE: BRIEFINGS AND DEBRIEFINGS

- Briefings are used in many settings, not just aviation (e.g., military medical, shift work). They quickly and accurately convey key aspects of a situation.
- Example for setting up a route in the Flight Management System (FMS)
  - "Three-step glass cockpit briefing" (Lutat & Swah, 2013)
    - "Build," load, or select the appropriate (instrument flight) procedure [route to follow] from the FMS database with direct reference to the current chart.
    - Rigorously check the FMS programming for compatibility with the aircraft clearance.
    - The pilot flying (workload permitting) briefs the procedure from the FMS, while pilot monitoring (ideally) checks the programming against the appropriate chart(s).
- What is a "debriefing"? (Forrest, 2015)
  - Discussion after completing a task. How did it go?
- Briefings in theory vs. briefings in practice (Chandra and Markunas, 2017)

CHECKLISTS

CHECKLISTS – CONCEPTS AND VOCABULARY

- Normal checklists are for checking, not to-do lists
- Abnormal/emergency checklists are READ-DO lists
- Electronic checklists
  - https://skybrary.aero/articles/checklists-purpose-and-use
- Checklists are used in many domains. Broad applications.

Higgins, W.Y., & Boorman, D.J. (2016) An Analysis of the Effectiveness of Checklists when combined with Other Processes, Methods and Tools to Reduce Risk in High Hazard Activities. *Boeing Technical Journal. (*Available in Canvas, Supplementary Readings)

NORMAL CREW CHECKLISTS

- Intended to verify/check actions that were accomplished from memory
  - Flow patterns
  - Mnemonics
    - e.g., GUMPS for pre-landing checklist – gas, undercarriage, (full rich fuel) mixture, propeller, seat belts & switches set
- On aircraft that require a crew of two, often the non-flying pilot will read the checklist and each item will be accomplished and acknowledged.
- Examples
  - Before start
  - Before Taxi/After Engine Start
  - Before takeoff, After takeoff
  - Pre-approach
  - Before landing

NON-NORMAL CREW CHECKLISTS

- Abnormal/Non-normal
- Emergency
- Quick Reference Handbook (physical book)
  - https://skybrary.aero/articles/quick-reference-handbook-qrh

- Can be part of an Electronic Checklist (ECL) shown on a flight deck or hand-held display

CHECKLISTS DESIGN ISSUES

- Typical user-interface issues
  - Content (correct steps, correct order etc.)
  - Clear communication (wording)
  - Format (easy to read, clear mapping of steps)
  - Help the pilot to ensure every necessary item is completed

TWO “BEFORE START” CHECKLISTS FOR B-757

757

BEFORE START

GEAR PINS ..... REMOVED
LOGBOOK ..... ON BOARD
OXYGEN ..... SET (B)
CABIN SIGNS ..... (_____), ON
FLT INSTRUMENTS ..... SET (B)
PARK BRAKE ..... SET
FUEL CONTROL ..... CUTOFF
FUEL ..... ____, CLRD WITH ____

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PAPERS ..... ON BOARD
AIRSPEED BUGS ..... SET (B)
TRIM ..... ____ UNITS & ZERO
FLIGHT CONTROLS ..... CHECKED

AFTER START

ENGINE ANTI-ICE ..... ____
ISOLATION SWITCH ..... OFF
RECALL ..... CHECKED
AUTOBRAKE (If installed) ..... RTO

TAXI

FLAPS ..... _____

Figure 6. *A Boeing B-757 checklist used by one airline.*

B-757/767
PILOT'S CHECKLIST

Full Functional Check Required on First Flight of the Day

- FIRE WARNING
- STANDBY POWER
- TCAS
- SELCAL

BEFORE START

C	■	EXTERIOR / INTERIOR PREFLIGHT	COMPLETE
C	■	LOGBOOK	CKD
C	■	CIRCUIT BREAKERS	CKD
C		TAKEOFF CONFIG WARNING	CKD
C		HYDRAULIC PANEL	SET
C		EEC SWITCHES	NORMAL / ON
C		YAW DAMPERS	ON
C		ANTISKID	ON
C	■	IRS MODE SELECTORS	NAV
C		VOICE RECORDER	CKD
C		ELECTRICAL PANEL	SET
C		EXTERIOR LIGHTS	SET
C		WING / ENGINE ANTI-ICE	OFF
C&F	■	FUEL REQUIRED	ONBOARD
C		FUEL PANEL	SET
C		ENGINE START PANEL	SET
		RAT SWITCH	OFF
		EMERGENCY LIGHTS	ARMED
		PASSENGER OXYGEN	OFF
	■	EQUIPMENT COOLING	NORMAL / AUTO
	■	CABIN ALTITUDE CONTROL	SET
	■	NO SMOKING / SEAT BELT SIGNS	AS REQD / ON
C		WINDOW HEAT	ON
C	■	(767) CARGO HEAT	ON
C		AIR CONDITIONING PANEL	SET
C	■	MODE CONTROL PANEL	SET
C&F	■	WINDOWS	CLOSED & LOCKED
A		OXY MASK / REG / INTPH	CKD
C&F		INSTRUMENT SOURCE SELECTORS	CKD
C&F	■	ALT / FLT & NAV INSTRUMENTS	SET / XCKD
C		RESERVE BRAKES (767 & STEERING)	[illegible]
C		STANDBY FLIGHT INSTRUMENTS	CKD
C		STANDBY ENGINE IND SELECTOR	AUTO
C		AUTO BRAKES	OFF
C		(ER) INSF HEADING REFERENCE SW	NORMAL
C		ALTN FLAPS (LE & TE)	NORMAL / OFF
C		LANDING GEAR	DOWN / 3 GREEN
C		ALTN GEAR EXTENSION	OFF
C		GND PROX OVRD SWITCHES	OFF
C		(ER) FLT INSTRUMENT BUS PWR	NORMAL
C	■	EICAS	CKD
C		SPEEDBRAKES	DOWN
C		STAB TRIM SWITCHES	NORMAL
C		THROTTLES	IDLE
C		FUEL CONTROL SWITCHES	CUTOFF
C		FLAPS	UP
	■	RADIOS / TRANS / TCAS / RADAR	CKD / SET
		FIRE WARNING	CKD
	■	PARKING BRAKE	AS REQD
	■	FLIGHT ATTENDANT BRIEFING	COMPLETE
	■	DEPARTURE BRIEFING	COMPLETE

When Paperwork Received

	■	ACARS INFO	ENTERED
F	■	FMS / RUNWAY DATA / TMS PANEL	[illegible]

PUSHBACK/START

F		DOOR LIGHTS	CKD
F		HYDRAULIC PANEL	SET
F		RED ANTI-COLLISION	ON
F		FUEL PANEL	SET
F		PACKS	OFF

After Start

F		AIR CONDITIONING PANEL	SET
C&F		ENGINE ANTI-ICE	AS REQD
F		APU	AS REQD
F		EICAS	RECALL
		AUTO BRAKES	RTO

TAXI

F		FLAPS	____ / ____
C&F		AIRSPEED BUGS	____ / XCKD
C&F		ALTIMETERS	____ / XCKD
N		FLIGHT / NAV INSTS	SET / XCKD
N		AIL / RUD / STAB TRIM	0 / 0 / 0
N		FLIGHT CONTROLS	TOPS / BOTTOMS

DELAYED START

F		APU	AS REQD
F		PACKS	OFF
F		ENGINE ANTI-ICE (BOTH)	OFF

After Start

F		AIR CONDITIONING PANEL	SET
C&F		ENGINE ANTI-ICE	AS REQD
F		APU	AS REQD
F		EICAS	RECALL

BEFORE TAKEOFF

C&F		FLAPS	____ / ____
F		FLIGHT ATTENDANTS	NOTIFIED / ACKD
F		FLT DECK DOOR / WINDOWS	CLSD & LCKD
F		SHOULDER HARNESS	ON
F		TAKEOFF BRIEFING	COMPLETE

Final Items

F		EXTERIOR LIGHTS	AS REQD
F		TRANS / TCAS	SET

FAA APPROVED
DATE 11 25 92

Figure 7. *A Boeing B-757 checklist used by another airline.*

*Degani, A. & Weiner, E.L. (1994) *On the Design of Flight Deck Procedures*. NASA Contractor Report 177642. https://ntrs.nasa.gov/citations/19940029437 (see pp. 19-20 for sample checklists)

CHECKLIST EVOLUTION: STANDARDIZATION VS. FLEXIBILITY

- Sample checklists for B-757 from two different airlines in Degani and Weiner (1994)
- Standardization is good…
- Over time, transitioned to using versions from the original equipment manufacturer
- Eventually, these versions in the Aircraft Flight Manual (AFM) were coded into electronic checklists
- However, it is difficult to modify after the checklists are part of the airplane software (due to Certification costs)

REFERENCES

https://www.faa.gov/pilots/training

https://www.faa.gov/regulations_policies/handbooks_manuals/aviation

Instrument flying handbook

AQP Program

AQP documents

https://simpleflying.com/level-d-simulator/

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DRIVING VS. FLYING

DIMENSIONS FOR COMPARING FLYING VS. DRIVING

Driving

- Private risk (non-commercial ops)
- 1 driver
- Regulation by State DOT
- Glances away (few seconds at most)
- Insurance industry

What else?

Airline (Commercial) Flying

- Public risk
- 2 pilots (airline ops)
- Federal regulator
- Can look away for much longer (cruise), (practiced) instrument scan

What else?

MANAGING RISK: OPERATIONAL COMPLEXITY IN ~~FLIGHT OPERATIONS –~~ DRIVING: THE "ODD"

ATC Intervention (such as)
- (Late) route amendments
- Unpublished restrictions
- Vectors

Operator Factors
- Independence vs. dependence on Dispatch
- Clarity and consistency of pilot roles in reviewing route

Operational Complexity
- ATC interventions
- ~~Aircraft~~ Vehicle equipment or performance
- ~~Crew~~ Driver factors
- Operator factors
- Environment factors

~~Aircraft~~ Vehicle Factors
- Lack or unreliability of automated systems
- Performance characteristics

Environment Factors
- Terrain
- Traffic
- Weather
- Prohibited ~~airspace~~ roads

~~Crew~~ Driver Factors
- Expectations (confirmation bias)
- Fatigue
- Communication style
- Distractions
- Local area familiarity
- Familiarity with different types of routes

(2/18/26 lecture)

MORE DRIVING VS. FLYING

Driving

- Relatively little training and currency checks
- Wider population of operators
- Could stop and reset/review
- 2-D task

How does the "Operational design domain" (ODD) apply to flying vs. driving?

Flying

- Intensive training and currency reviews
- Relatively limited set of operators
- Certifications and approval of both pilots and aircraft
- ATC monitors flights and can aid in emergencies
- Always in motion
- Public expects high safety levels for commercial operations
- 3-D task